Question 2.

Date: / / Mon Tue Wed Thu Fri Sat Sun

$$\vec{a}^{(3)} = W^{(3)}\vec{a}^{(2)} + \vec{b}^{(3)}$$

$$\vec{a}^{(2)} = W^{(2)}\vec{a}^{(1)} + \vec{b}^{(2)}$$

$$\vec{a}^{(1)} = W^{(1)}\vec{a}^{(0)} + \vec{b}^{(1)}$$

$$(a) = \vec{a}^{(3)} = W^{(3)}\left(W^{(2)}\vec{a}^{(1)} + \vec{b}^{(2)}\right) + \vec{b}^{(3)}$$

$$= W^{(3)}\left\{W^{(2)}\left(W^{(1)}\vec{a}^{(0)} + \vec{b}^{(1)}\right) + \vec{b}^{(2)}\right\} + \vec{b}^{(3)}$$

$$= W^{(3)}\left[W^{(2)}W^{(1)}\vec{a}^{(0)} + W^{(2)}\vec{b}^{(1)} + \vec{b}^{(2)}\right] + \vec{b}^{(3)}$$

$$= W^{(3)}W^{(2)}W^{(1)}\vec{a}^{(0)} + \underbrace{W^{(3)}W^{(2)}}_{6\times 6}\vec{b}^{(1)} + W^{(3)}\vec{b}^{(2)} + \vec{b}^{(3)}$$

$$(b) = \tilde{a}^{(1)} = \tilde{W}\vec{a}^{(0)} + \tilde{b}$$

so: $\tilde{W} = W^{(3)}W^{(2)}W^{(1)}$

$$\tilde{b} = W^{(3)}W^{(2)}\vec{b}^{(1)} + W^{(3)}\vec{b}^{(2)} + \vec{b}^{(3)}$$